

2 The photograph shows a pollen tube growing in a culture medium on a glass slide.



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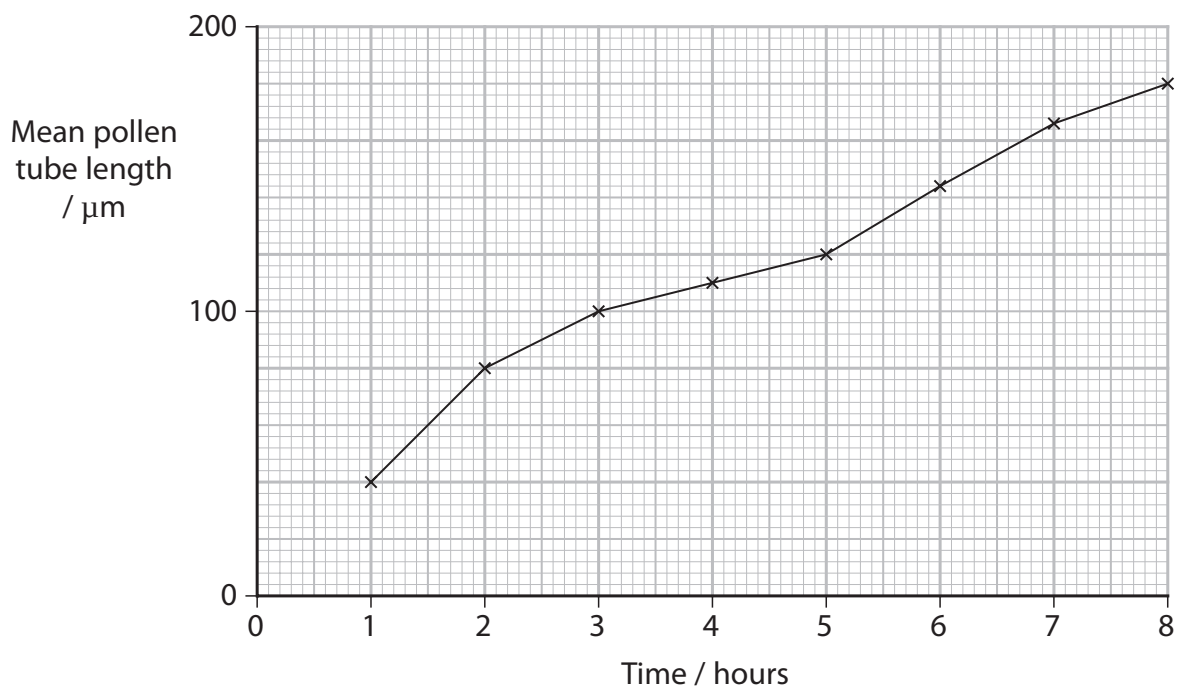
Pollen grains were placed on glass slides covered in growth medium.

Eight slides were incubated at 24°C.

After one hour, one slide was placed under a microscope and the lengths of 10 pollen tubes were measured.

The measurements were repeated at hourly intervals with the remaining seven slides.

The graph shows the results.



- (a) Calculate the mean rate of growth of the pollen tubes between one and five hours.

Include units with your answer.

(2)

Answer

(b) Explain how changes in temperature could affect the rate of growth of pollen tubes.

(2)

(c) Abiotic variables were controlled in this investigation.

(i) State **two** abiotic variables other than temperature that could affect this investigation.

(2)

First variable

Second variable

(ii) Choose **one** of the variables you have identified in (i).

Describe how this variable can be controlled.

(1)

Variable

Method of control

(iii) Suggest what effect not controlling this variable would have on the results.

(1)

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(d) Describe how this method can be modified to investigate the effect of age of pollen on the rate of growth of pollen tubes.

(2)

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(Total for Question 2 = 10 marks)